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Clue before correction: ChatGPT-enhanced strategy for promoting autonomous and reflective language learning

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ABSTRACT

Purpose: This study presents a generative AI-supported revision task designed to promote adaptive learning, learner autonomy, and metacognitive engagement in second language (L2) writing through clue-based interaction with ChatGPT. Unlike traditional chatbot use for direct error correction, which positions learners as passive recipients, this activity was intentionally structured to require active problem-solving, prompting students to infer correct solutions from guided hints.

Design/methodology/approach: Fifty-eight university students (CEFR A1–B1) completed a bilingual activity in English or Spanish that used indirect prompts to guide self-revision, supported by ChatGPT. Learner–AI interaction transcripts and open-ended reflections were analyzed using Cognitive Load Theory, formative feedback models, and the newly developed ICAP-ME framework, which extends the ICAP model to include metacognitive and affective dimensions.

Findings: ChatGPT provided structured, adaptive feedback that reduced cognitive load and supported personalized revision pathways, particularly for more proficient learners. Learner prompting behavior strongly influenced feedback quality and depth.

Originality/value: This study introduces (1) a replicable instructional design for GenAI-supported revision, (2) the ICAP-ME pedagogical framework for analyzing AI-mediated learning, and (3) design principles for equitable, developmentally appropriate use of GenAI in language education. It contributes to emerging research on how AI tools can foster autonomy, reflective engagement, and strategic learning.

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1. Introduction

As generative AI tools like ChatGPT become more common in education, they offer adaptive, personalized, and immediate support across learning environments (Delgado et al. 2020; Johnson 2019; Wei 2025). In language education, they facilitate real-time feedback,

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scaffold learning, and provide individualized language modeling. These features are particularly valuable for learners of English as a Foreign Language (Panagiotidis 2024; Wang 2022).

Research shows that GenAI can enhance engagement, reduce anxiety, and increase motivation by offering personalized, low-stakes practice (Biju et al. 2024; Du and Alm 2024; Wei 2023). However, introducing GenAI with teacher guidance is crucial to addressing concerns about academic dishonesty that students might otherwise develop (Elsayed et al. 2024; Jo 2024). At the same time, integrating GenAI raises significant pedagogical and ethical concerns. One challenge is that students may rely too heavily on AI-generated content, thereby avoiding critical thinking and deeper linguistic processing (Akgun and Greenhow 2022; Fyfe 2023). This reliance can weaken learner autonomy and hinder the development of essential writing and revision skills (Kim et al. 2025; Xiao and Zhi 2023). Instructors also face difficulties evaluating authentic student performance, given the limitations of current AI-detection tools (Perkins et al. 2024). Although researchers have examined classroom perceptions of GenAI (Chu and Wang 2025; Mohamed 2024) and proposed task design strategies (Lin, Chang, and Winne 2025; Liu and Xiao 2025), there is still a need for robust frameworks that support ethical and pedagogically sound integration.

This study addresses these challenges by presenting a task where students prompt ChatGPT to provide indirect clues rather than direct corrections. This encourages learners to identify and revise their own errors, promoting agency, strategic thinking, and reflective learning (Esfandiari and Allaf-Akbary 2024; Sawangwan 2024; Xiao and Zhi 2023; Zhao 2025). By positioning ChatGPT as a formative partner, the design emphasizes learner-driven personalization shaped by prompting choices and cognitive engagement.

Students used the free version of ChatGPT to ensure equitable access. The activity was implemented in English and Spanish to examine whether engagement and feedback quality differed by language. While earlier versions of ChatGPT favored English (Brown et al. 2020; Seghier 2023), this study found no major differences between the two languages.

In addition to its pedagogical focus, this study introduces a unique methodological affordance: the ability to analyze full transcripts of learner-ChatGPT interactions. Unlike traditional classrooms, where engagement is inferred from external behaviors or reflections, GenAI-mediated tasks provide a granular, turn-by-turn record of learners' decisions, questions, revisions, and emotional responses. This allows researchers to observe subtle cognitive, metacognitive, and affective shifts often invisible in other contexts. Moreover, since ChatGPT is perceived as a non-judgmental, informal partner (Fakour and Imani 2025), students express themselves more authentically, especially in digital contexts where younger learners are more open to showing uncertainty and emotion.

However, interactional patterns such as strategic prompting, delayed doubt, and affective responses are not fully captured by traditional models of engagement. The ICAP framework (Interactive, Constructive, Active, Passive; Chi and Wylie 2014) provides a foundation for understanding learner engagement but focuses mainly on observable behavior. This study therefore introduces ICAP-ME, an expanded framework that includes metacognitive and affective dimensions of engagement in GenAI-supported learning contexts. Grounded in research on metacognitive engagement and learner-centered pedagogy, ICAP-ME informs the design of AI-integrated strategies that foster

personalization, critical thinking, and autonomy. This study contributes to ongoing efforts to integrate GenAI ethically and effectively into language education.

2. Literature review

2.1. Error-based and reflective learning

The pedagogical design of this study draws on research in error-based learning, cognitive engagement theory, and adaptive feedback. The errorful generation principle posits that making and correcting errors enhances memory, conceptual understanding, and motivation (Kornell, Hays, and Bjork 2009; Potts et al. 2019). This effect is strongest when learners perceive a reasonable chance of success, which increases curiosity, attention, and sustained cognitive effort (Kang, McDermott, and Roediger 2007; Seabrooke et al. 2019). These motivational effects underpin the errorful generation effect, which suggests that incorrect guesses followed by feedback can result in deeper learning than passive study (Potts and Shanks 2014).

Zhang and Fiorella's (2022) productive failure model formalizes this process through two phases: Generating Errors (GE) and Detecting and Correcting Errors (DCE). The current study operationalizes this model by prompting learners to seek subtle cues from ChatGPT instead of direct corrections. This design encourages hypothesis testing, reasoning, and reflection, while also eliciting emotional engagement, factors known to enhance memory and attention (Brod, Hasselhorn, and Bunge 2018).

In contrast, traditional peer correction often leads to disengagement and reduced accountability. Tulis (2013) called this the 'Bermuda Triangle of Error Correction,' where original authors are removed from revision. ChatGPT instead offers personalized, peer-like scaffolding that fosters agency, reflection, and iterative improvement, supporting confidence, resilience, and strategic awareness (Tulis, Steuer, and Dresel 2017; Young and Shishido 2023).

2.2. Formative assessment and learner autonomy

The activity aligns with research on formative assessment, defined by Ramaprasad (1983) and elaborated by Black and Wiliam (2009) as feedback that clarifies goals, provides insight into progress, and guides learners toward improvement. Following Hattie and Timperley's (2007) framework, effective feedback addresses three questions: *Where am I going?* (goal clarity), *How am I going?* (performance feedback), and *Where to next?* (strategic guidance). In their model, these functions correspond to four levels of feedback: task-level feedback, which focuses on accuracy; process-level feedback, which targets strategies and patterns; self-regulation feedback, which supports learners' monitoring and adjustments; and self-level feedback, which involves praise or personal evaluation and is considered least effective.

This study applies this framework because it provides a robust, evidence-based method for distinguishing forms of feedback that meaningfully support learning from those that do not, and it aligns with the activity's focus on accuracy, strategic reasoning, and learner self-regulation. Accordingly, RQ2 draws directly on this model to examine the

extent to which ChatGPT's feedback reflects these established categories and supports learning-relevant processes.

Importantly, chat-based hints can foster learners' own error-detection skills, enabling them to generate self-feedback aimed at reaching communicative or linguistic goals – an especially powerful process when learners have enough prior knowledge to analyze their output and adjust strategies. By prompting students to notice discrepancies, seek better alternatives, or gather additional information, ChatGPT promotes precisely the kind of strategic engagement that Hattie and Timperley identify as central to effective learning.

Research further shows that elaborated feedback, feedback that provides explanatory information, cues, or strategic guidance rather than simple correctness judgments, produces significantly higher learning gains than verification-only feedback (Hattie and Timperley 2007; Shute 2008), yet time constraints often prevent teachers from providing such clarification to every student. Chat-based interaction addresses this limitation by offering immediate, personalized clarification that requires learners to actively process and respond to the information. The timing of feedback is also significant: while some tasks benefit from immediate feedback and others from delayed feedback, the present activity allows learners to choose when to request support, accommodating individual preferences. This is particularly relevant given that students tend to seek help only when confident, often avoiding situations where errors are visible, such avoidance limits opportunities for learning from mistakes.

Hattie and Timperley emphasize that feedback is one of the most influential factors in learning, but its effectiveness depends on conditions such as clarity, purposefulness, relevance to prior knowledge, low threat, and alignment with specific learning goals. Feedback should not be directed at the self but at the task, the processes used, or the learner's regulation of those processes. A crucial distinction is that this activity therefore avoids self-level praise, which refers to feedback directed at the learner as a person (e.g. 'You are a good writer') rather than at the task or process. Research shows that such self-level feedback offers little instructional value and can reduce motivation by promoting fixed rather than growth mindsets (Dweck 2006; Hattie and Timperley 2007). Chat-based feedback in this activity meets these criteria by offering low-threat, goal-oriented guidance that promotes active cognitive engagement, self-assessment, and iterative improvement, conditions that support learning in and beyond the classroom.

The design also promotes autonomous learning, understood as learners' capacity to take responsibility for their own learning through decision-making, goal setting, and self-assessment (Benson 2011; Holec 1981; Little 1991). Autonomy in this sense is not mere independence, but a dynamic process of self-regulation and reflection, often exercised within social and pedagogical contexts that encourage collaboration and dialogue (Reinders and White 2016). Recent work on language learning beyond the classroom (LBC) conceptualizes autonomy in terms of locus of control, the extent to which learners, rather than teachers or materials, direct their own learning trajectories across different settings (Benson 2011; Reinders and Benson 2017). From this perspective, tools such as ChatGPT can be understood as resources within a broader ecology of LBC: learners choose when, how, and for what purposes to engage with them, and thereby exercise autonomy by configuring their own learning environments and trajectories. The activity therefore situates autonomy as both a personal and interactive capability, developed through learners'

active engagement with digital tools and their ability to shape how these tools support their learning inside and beyond the classroom.

Within the framework of Self-Determination Theory (SDT) (Deci and Ryan 2000), autonomy is understood as one of three innate psychological needs, alongside competence and relatedness that underpin intrinsic motivation and sustained engagement. When these needs are supported, learners are more likely to experience a sense of volition, purpose, and connection in their learning processes. The activity supports these needs by granting learners control over the timing, pacing, and depth of feedback, thereby nurturing a sense of competence through personalized challenge, autonomy through choice, and relatedness through meaningful interaction with a responsive system. In line with LBC research that reconceptualizes the classroom as one node in a wider network of learning opportunities (Reinders and Benson 2017), the design encourages learners to appropriate the feedback process in ways that can be transferred to self-initiated learning beyond formal instruction.

This learner-controlled approach aligns with research showing that opportunities to self-direct feedback processes can reduce anxiety, enhance confidence, and promote deeper cognitive engagement (Hattie and Timperley 2007; Hsu and Chen 2025). It also resonates with descriptions of *self-directed naturalistic learning* in LBC, where learners deliberately set up situations for learning but shift their attention toward meaningful communication or content once engaged (Benson 2011; Reinders and Benson 2017). By enabling learners to initiate, tailor, and interpret feedback interactions with ChatGPT, the activity helps them develop the strategic and reflective capacities needed to manage similar learning opportunities in out-of-class, technology-mediated contexts. In this sense, the design operationalizes theoretical principles of autonomy, self-determination, and LBC, positioning learners as active agents in their development rather than passive recipients of information.

2.3. Cognitive load theory, human cognitive architecture, and sociocognitive foundations

Cognitive Load Theory (CLT; Sweller, Ayres, and Kalyuga 2011) and Human Cognitive Architecture (HCA; Kirschner, Sweller, and Clark 2006) form the theoretical basis for the instructional design evaluated in RQ1. Both frameworks emphasize that working memory is severely limited when dealing with unfamiliar information, while long-term memory is the primary site of learning, where knowledge becomes organized into schemas that allow fluent and automatic performance. Instruction must therefore be designed to minimize unnecessary cognitive demands, support schema acquisition, and align task complexity with the learner's level of expertise.

CLT differentiates between intrinsic load (the inherent complexity of the material), extraneous load (inefficiencies caused by poor instructional design), and germane load (productive cognitive effort devoted to learning). Novice learners are especially vulnerable to overload when required to search for errors or solutions without sufficient guidance. Kirschner, Sweller, and Clark (2006) argue that minimally guided or discovery-based instruction often forces novices into unproductive problem-space searches, consuming working memory without contributing to schema development. Guided instruction with well-designed prompts is therefore essential for enabling learners to focus on the most relevant features of language form and meaning.

The ChatGPT-supported revision activity operationalizes these principles by providing small, sequential clues rather than broad or vague prompts, reducing extraneous cognitive load and preventing inefficient search behavior. At the same time, targeted hints activate prior knowledge and promote long-term memory retrieval, which strengthens schema construction. This aligns with evidence that learning improves when students receive structured scaffolding that fades gradually as expertise increases, allowing learners to assume greater strategic control.

The design also draws on sociocognitive perspectives on metacognition, which conceptualize learning as a process emerging through reflection, explanation, and dialogic interaction (Chi and Wylie 2014). Metacognitive development is triggered when learners compare their initial hypotheses with corrective cues, experience cognitive conflict, and articulate the reasoning behind revisions. ChatGPT's dialogic prompts foster this process by encouraging students to explain errors, justify choices, and anticipate alternative solutions, behaviors central to self-regulated learning.

Together, these theoretical strands inform the five analytical dimensions used in RQ1. Cognitive Load Management (CLM) reflects CLT's emphasis on avoiding overload through focused, sequential guidance. Long-Term Memory Activation (LMA) corresponds to the need to cue and strengthen relevant schemas. Scaffolding (SCF) captures the guided-to-independent progression essential for novice learners. Rule Consolidation (RC) represents opportunities for reinforcing grammatical principles after self-correction. Metacognitive Engagement (ME) draws directly from sociocognitive theories of reflection and dialogue that support learners in monitoring and regulating their performance.

This theoretical grounding clarifies how the ChatGPT-supported activity was designed to optimize cognitive resources, promote durable learning, and facilitate reflective engagement, providing the foundation for the evaluative rubric used in Section 4.4.1.

3. Extending the ICAP framework for AI-mediated learning: the ICAP-ME model

3.1. The ICAP framework

The ICAP framework (Chi and Wylie 2014) conceptualizes cognitive engagement along a four-level hierarchy: Passive, Active, Constructive, and Interactive, reflecting progressively deeper forms of learning. Passive engagement involves minimal processing (e.g. listening or reading), whereas Active engagement includes manipulation or simple elaboration (e.g. repeating, highlighting, short responses). Constructive engagement requires learners to generate new ideas, explanations, or inferences, and Interactive engagement involves reciprocal co-construction of understanding with another agent. ICAP has been validated extensively across STEM instruction, writing studies, collaborative problem solving, and intelligent tutoring systems (Chi 2009; Fiorella and Mayer 2016; Menekse et al. 2013; Nokes-Malach, Richey, and Gadgil 2015; VanLehn 2011).

3.2. Limitations of ICAP in AI-mediated contexts

Although the ICAP framework offers a robust taxonomy of observable engagement (Chi and Wylie 2014), several limitations reduce its suitability for analyzing AI-mediated

learning. First, ICAP is behaviorally reductionist: it classifies engagement solely through visible actions, overlooking covert processes such as reflection, uncertainty monitoring, trust calibration, and strategy planning, processes that are central to how learners evaluate and regulate AI-generated feedback (Fiorella 2020; Winne 2018).

Second, ICAP's categories can be ambiguous, particularly between Active and Constructive engagement, since similar surface behaviors (e.g. asking a question) may reflect different cognitive operations. Researchers report that this ambiguity complicates coding reliability and interpretive validity in dialogue-based environments (Greene, Sandoval, and Bråten 2016; Kirschner et al. 2018).

Third, ICAP does not incorporate affective or motivational dimensions, despite extensive evidence that emotions such as curiosity, frustration, confusion, and enjoyment shape cognitive load, persistence, and depth of processing (D'Mello and Graesser 2012). These emotional processes are especially salient in AI-mediated contexts where system responses may be unpredictable, ambiguous, or incorrect.

Finally, ICAP is not designed to capture AI-specific behaviors that are increasingly documented in empirical studies of learner-AI interaction, including iterative prompt refinement, skepticism toward machine reasoning, meta-dialogic evaluation, and emotional regulation in response to system errors (Holmes, Bialik, and Fadel 2019; Kasneci et al. 2023; Thurn et al. 2023). These behaviors are essential components of digital literacy and self-regulated learning in generative AI environments, yet they fall outside ICAP's behavioral hierarchy.

Taken together, these limitations suggest that while ICAP remains a valuable theoretical foundation, it requires extension to account for the metacognitive and affective dynamics characteristic of AI-mediated learning.

3.3. Conceptual contribution of ICAP-ME

The ICAP-ME model extends Chi and Wylie's (2014) ICAP framework by adding two dimensions essential for understanding engagement in AI-mediated learning: Metacognitive engagement (M) and Affective engagement (E). Rather than replacing ICAP's behavioral hierarchy (Passive, Active, Constructive, Interactive), ICAP-ME conceptualizes engagement as a layered system in which observable behaviors co-occur with internal regulatory and emotional processes (Figure 1).

Metacognitive dimension

The metacognitive component captures regulatory behaviors that frequently appear in learner-AI interactions but are invisible in ICAP's behavioral categories. Guided by self-regulated learning research (Azevedo and Cromley 2004; Winne 2018), this dimension includes five metacognitive categories:

- **Verification Seeking (VS)** – checking correctness, asking for confirmation, evaluating alternatives
- **Error Awareness (EA)** – noticing, acknowledging, or analyzing one's own errors
- **Trust Justification (TJ)** – questioning, evaluating, or defending the validity of AI feedback

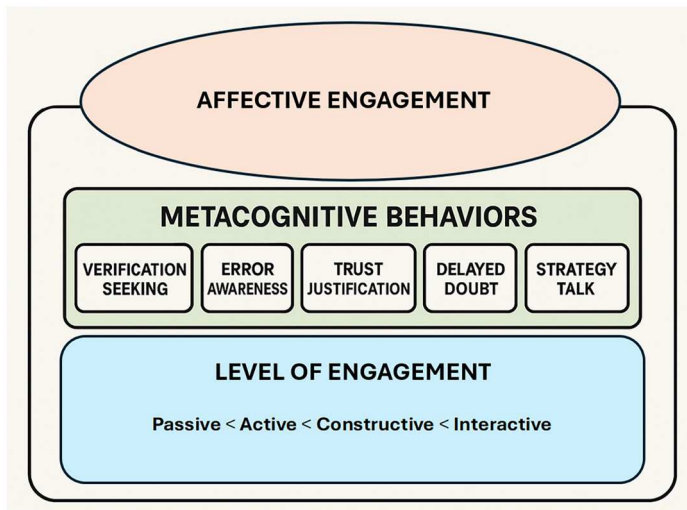


Figure 1. The ICAP-ME Pedagogical Framework: A Model of Cognitive, Metacognitive, and Affective Engagement in GenAI-Supported Learning.

- **Delayed Doubt (DD)** – revisiting or reconsidering previous responses after reflection
- **Strategy Talk (ST)** – articulating planning, reasoning, or choice of approach

Each metacognitive behavior can occur at different depths of behavioral engagement along the ICAP continuum. For example, Verification Seeking may be Passive (‘Is this correct?’) or Constructive (‘Why is this wrong?’), depending on the cognitive demand and the nature of the learner’s contribution.

Affective dimension

The affective dimension captures emotional reactions such as curiosity, frustration, confusion, relief, or enjoyment that shape engagement depth, cognitive load, and persistence (D’Mello and Graesser 2012). Affective engagement is coded separately from behavioral and metacognitive categories because emotional expressions can occur at any behavioral level and are analytically distinct.

Theoretical rationale

Integrating metacognitive and affective dimensions:

1. Addresses ICAP’s behavioral bias by incorporating covert regulatory and emotional processes.
2. Reflects sociocognitive theories that view learning as emerging through reasoning, contradiction, and reflection (Black and Wiliam 2009; White and Frederiksen 1998).
3. Aligns with Cognitive Load Theory, acknowledging that metacognition and emotion influence intrinsic, extraneous, and germane load (Kirschner, Sweller, and Clark 2006; Sweller, Ayres, and Kalyuga 2011).
4. Captures AI-specific engagement patterns, such as doubt, trust calibration, prompt refinement, and ongoing reflection beyond the immediate task.

Together, these additions allow ICAP-ME to provide a more comprehensive account of learner engagement in complex, AI-supported environments.

3.4. Development of the ICAP-ME framework

The ICAP-ME model was developed through an iterative process integrating theoretical synthesis and empirical analysis of learner-AI dialogues.

Phase 1: theoretical synthesis

A multidisciplinary literature review on metacognition, affect, cognitive load, error-based learning, and AI-mediated writing (Flavell 1979; Holmes, Bialik, and Fadel 2019; Sweller, Ayres, and Kalyuga 2011) highlighted types of engagement missing from the original ICAP categories.

Phase 2: exploratory transcript coding

Pilot coding of ChatGPT-learner transcripts revealed recurrent behaviors such as confirmation seeking, error analysis, trust evaluation, delayed doubt, and explicit planning. These patterns aligned with established SRL constructs but were not accommodated by ICAP alone, supporting the need for an extended framework.

Phase 3: category construction

Five metacognitive categories (VS, EA, TJ, DD, ST) were defined, each mapped across the ICAP continuum (Passive–Interactive). A separate affective category was created to capture emotional expressions identifiable through transcript cues and reflective comments.

Phase 4: consensus refinement

Two coders independently applied the emerging framework. Average percentage agreement was 80.25%. Discrepancies were resolved using consensual qualitative analysis (Hill, Thompson, and Williams 1997), clarifying category boundaries and strengthening reliability.

Phase 5: finalization

The final ICAP-ME matrix integrates:

- ICAP behavioral engagement
- the five metacognitive categories distributed along ICAP levels
- affective engagement coded independently

This structure allows nuanced analysis of learner-AI interaction that accounts for cognitive, regulatory, and emotional dimensions.

3.5. Methodological justification

AI-mediated writing and revision involve dynamic interplay between behavioral, metacognitive, and emotional processes that ICAP alone cannot fully capture. Learners may

seek verification, identify errors, question AI feedback, justify trust, refine prompts, regulate frustration, or return to earlier decisions after reflection. These processes are central to digital literacy, feedback uptake, and self-regulated learning.

The ICAP-ME framework captures this complexity by integrating:

- Behavioral engagement (ICAP) – what the learner does overtly
- Metacognitive engagement (M) – how the learner monitors, evaluates, and plans
- Affective engagement (E) – how the learner responds emotionally to the task and the AI

By analyzing these dimensions, ICAP-ME provides a theoretically grounded and pedagogically relevant account of engagement in contemporary AI-mediated learning environments. It also provides the analytic foundation used in this study to evaluate student engagement (RQ3).

Research questions

1. How does the design of the ChatGPT-supported revision activity align with principles from Cognitive Load Theory and Human Cognitive Architecture?
2. How does the feedback generated by ChatGPT support personalized, task-level, process-level, and self-regulatory learning according to established feedback models?
3. What levels and types of student engagement are evident during the activity, particularly in terms of individualized interaction and adaptive feedback, based on transcript data and post-task reflections coded with a ICAP-ME framework?

4. Methods

4.1. Participants

The study involved 58 university students who voluntarily participated from English and Spanish language modules. Based on a diagnostic test, participants were grouped into CEFR levels (A1 = 27, A2 = 17, B1 = 14), reflecting their post-hoc proficiency. Including beginners was essential, as autonomous learning is particularly challenging for those lacking foundational language skills and confidence in decision-making. However, Eurostat (2024) reports that English instruction begins at the primary level across the EU, making true beginners in English virtually absent at both the secondary and university levels. As a result, the A1 group included only Spanish learners. While prior exposure to generative AI tools varied, all participants were familiar with ChatGPT. The study focused on non-language majors aged 19–25, with balanced gender representation and diverse academic backgrounds.

4.2. Instructional procedure

Learners completed a six-step revision activity designed to foster metacognitive engagement through AI-generated feedback (see Appendix A). Using standardized prompts in either English or Spanish, they submitted self-written texts to ChatGPT and requested indirect, clue-based assistance rather than direct corrections. All subsequent steps,

including iterative revision, evaluation of AI suggestions, and strategy reflection, took place during the ongoing conversation with ChatGPT.

Detailed step-by-step instructions given to students and a comprehensive outline of the prompt structure and task sequence are provided in Appendices A and B, respectively. These appendices offer in-depth information essential for understanding and replicating the student–AI interaction process.

University participants completed the task independently at home, outside class hours, using the free version of ChatGPT to ensure uniform access and eliminate disparities related to subscription tiers or device capabilities. No external assistance was permitted. After completing the activity, students submitted a post-task written reflection. This reflection was intentionally open-ended, as we aimed to capture unprompted learner interpretations and emotional responses. We deliberately avoided using guiding questions, since we were interested in students' immediate reactions and did not want to influence their responses or prompt them to invent ideas they had not genuinely experienced during the activity. To ensure that the reflection was not limited by the students' language proficiency, they were allowed to write it in English, Spanish, or their native language.

4.3. Data sources

Two primary data sources were collected and paired for each university participant:

1. Student-ChatGPT transcripts (n = 58), recording all learner-AI interactions during the revision task.
2. Post-task written reflections, in which students described their revision strategies, feedback interpretation, and emotional responses.

4.4. Analytical frameworks

To explore the pedagogical effectiveness of the activity, we conducted three complementary analyzes. The first two analyzes focus on how ChatGPT functions as an instructional tool, examining its alignment with instructional design principles (RQ1) and the quality of its feedback (RQ2). The third analysis investigates how students respond to ChatGPT's feedback and scaffolding, using engagement data from transcripts and post-task reflections (RQ3). This layered approach allows us to examine not only the technological affordances of ChatGPT but also the (meta)cognitive and affective responses of learners engaging with it.

Three rubric-based analytical frameworks were employed to investigate the research questions that align with established cognitive and educational theories. A complete example of our analytical coding approach, applied to a student–ChatGPT interaction transcript, is available in Appendix C. This appendix illustrates the detailed interactional moves and coding categories utilized in the present study.

4.4.1. Evaluation of instructional design (RQ1)

A thematic content analysis of the transcripts was conducted using a rubric informed by Cognitive Load Theory (CLT) and Human Cognitive Architecture (HCA). The rubric assessed five dimensions (Table 1):

Table 1. Cognitive architecture-based rubric.

Code	Dimension	Description
CLM	Cognitive Load Management	Whether feedback reduced overload by offering focused, sequential clues.
LMA	Long-Term Memory Activation	Whether cues activated prior knowledge or reinforced grammar rules.
SCF	Scaffolding	Presence of guided prompts and gradual fading to support autonomy.
RC	Rule Consolidation	Evidence of grammar rule reinforcement after self-correction.
ME	Metacognitive Engagement	Prompts that encouraged reflection, clarification, or error anticipation.

Each transcript was analyzed at the interactional turn level to identify where and how these elements emerged.

4.4.2. Evaluation of feedback types (RQ2)

ChatGPT's feedback was evaluated using Hattie and Timperley's (2007) model of formative feedback, which categorizes responses according to their function:

Frequency and depth of each feedback type were recorded and compared across proficiency groups to evaluate adaptiveness (Table 2).

4.4.3. Evaluation of student engagement (RQ3)

Learner engagement was assessed using the ICAP-ME Pedagogical Framework, an extension of Chi and Wylie's (2014) model that includes metacognitive and affective dimensions. Table 3 defines each engagement type and provides examples. Five metacognitive behaviors were coded along the ICAP continuum (Passive, Active, Constructive, Interactive), and one affective dimension was coded separately, resulting in a matrix of engagement types (Table 4).

Student engagement was analyzed through qualitative coding of ChatGPT interaction transcripts and post-task reflections. Each engagement move, such as a clarification question, revision, or emotional expression, was coded individually to preserve analytical detail. We did not quantify frequencies because the goal of the analysis was interpretive rather than statistical. Raw counts would have been misleading due to differences in task length, learner confidence, types of errors, and time taken to reach correct responses. Instead, we evaluated each transcript as a whole to understand how the interaction unfolded. These evaluations were then used to identify recurring engagement patterns by proficiency level.

Coding validation. Two researchers independently coded all transcripts and reflections using the established rubrics. Discrepancies were discussed and resolved through a consensus-building process with a third researcher serving as an auditor. This approach followed principles of investigator triangulation and consensual qualitative research (Hill, Thompson, and Williams 1997), increasing the credibility and interpretive validity of the findings.

Table 2. Feedback coding rubric (Hattie & timperley framework).

Code	Feedback Level	Focus	Feedback Function
FTL	Task-Level	Accuracy, rule explanation, confirmation	"How am I going?"
FPL	Process-Level	Strategies, error patterns, alternatives	"Where to next?"
FSR	Self-Regulation	Prompts for reflection or self-assessment	Strategic goal-setting
FSL	Self-Level	Praise or unrelated personal judgment	Not instructional

Table 3. ICAP-ME pedagogical framework.

Metacognitive Category	Generic Description	Passive (P)	Active (A)	Constructive (C)	Interactive (I)
Verification Seeking (VS)	Learner checks correctness of their responses or understanding, often seeking confirmation or clarification.	'Is it correct?' (rote)	Applies a correction, asks again	Asks 'Why is this wrong?', evaluates alternatives	Negotiates correctness, builds on mutual understanding
Error Awareness (EA)	Learner identifies, acknowledges, or analyzes their own linguistic errors or misunderstandings.	Notes mistake without action	Fixes error after noticing it	Explains why it's wrong, challenges ChatGPT	Debates error with ChatGPT, co-constructs correction
Trust Justification (TJ)	Learner questions, evaluates, or defends the validity of ChatGPT's feedback or reasoning.	Accepts or rejects without reason	Seeks evidence ('I'll look it up')	Questions ChatGPT's reasoning: 'Why do you think that?'	Questions + offers sources or counterpoints
Delayed Doubt (DD)	Learner revisits or reconsiders a previous decision or response after reflection or later realization.	Silent revision later	Revisits earlier choice	Reflects: 'Now I think it was wrong because ...'	Re-engages ChatGPT to revisit earlier assumption
Strategy Talk (ST)	Learner articulates or reflects on their own learning strategies, problem-solving approaches, or planning processes.	Mentions effort vaguely	"Let me try this again"	'I read the rules first, then tried this.'	Plans aloud, adjusts strategy in collaboration with ChatGPT

Table 4. Affective engagement (e.g. expressions of curiosity, frustration, or enjoyment) was coded separately based on transcript cues and reflection content.

Affective Engagement (AF)	Learner expresses emotional, motivational, or attitudinal response to the task or the AI interaction	Humor, emojis  , exclamations ('ugh', 'yay', 'cool'), stated enjoyment or frustration	'I liked this', 'It was frustrating', 'I'll use it again', 'This surprised me'
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5. Results

This section presents findings from the three research questions, which focused on (1) the instructional design of the ChatGPT-supported revision activity, (2) the pedagogical quality of feedback generated by the AI, and (3) the types and levels of learner engagement observed across proficiency levels. These findings are drawn from paired analyzes of interaction transcripts and post-task reflections.

5.1. ChatGPT as an instructional partner: feedback and scaffolding functions

The first two analyzes examined how ChatGPT functioned within the instructional design of the activity, with a focus on its ability to scaffold learning and deliver pedagogically meaningful feedback.

RQ1: alignment with cognitive load theory and human cognitive architecture

The ChatGPT-supported revision activity demonstrated strong theoretical alignment with Cognitive Load Theory (CLT) and Human Cognitive Architecture (HCA). Analysis of learner-

AI transcripts revealed several ways in which the design supported effective cognitive processing, though patterns varied across CEFR proficiency levels.

Cognitive load management (CLM). ChatGPT delivered chunked, sequential feedback rather than full-text corrections across all groups, thereby reducing extraneous cognitive load. Importantly, the size and complexity of these segments varied systematically according to proficiency. For B1 learners, feedback often addressed broad conceptual categories such as the prompt ‘Here are some hints: 1. Title ... 2. Tense consistency ... 3. Prepositions ...’, which supported the processing of global linguistic patterns like narrative cohesion and tense alignment. A2 learners received more fine-grained, clause-level guidance, including prompts such as ‘depends on where I want to go’, which directed attention to specific grammatical structures while maintaining manageable cognitive demands. In the A1 transcript, ChatGPT segmented feedback into very small, concrete items, for instance: ‘el vivo en Prague.’ (‘he lives in Prague’), which helped to minimize processing load for beginners with limited linguistic resources.

Long-term memory activation (LMA). Evidence of LMA also varied by proficiency. B1 learners were regularly prompted to retrieve abstract grammatical knowledge, as in the explanation that ‘If you want to express that you still haven’t seen someone up to the present moment, the present perfect is correct ... but in a past narrative you’d use past perfect.’ A2 learners received activation cues focusing on core grammatical rules, such as the use of the subjunctive after *depende de*: ‘En español, después de ‘depende de ...’ cuando hablamos de una situación hipotética usamos el subjuntivo’ (‘In Spanish, after ‘depends on ...’, when referring to something hypothetical, we use the subjunctive’). In contrast, A1 learners were prompted to recall far more basic knowledge, for example: ‘La letra ‘ñ’ se usa en checo, pero en español usamos ‘ñ’, como en ‘niño’ (‘The letter ‘ñ’ is used in Czech, but in Spanish we use ‘ñ’, as in ‘niño’). These patterns indicate that retrieval-based learning opportunities were more prevalent at higher proficiency levels, while beginners received more declarative information due to limited schematic knowledge available for retrieval.

Scaffolding and fading (SCF). Evidence of instructional scaffolding was strongest in the B1 group. Learners frequently engaged in scaffolded self-correction supported by gradually fading guidance, as in the prompt: ‘Try to spot these issues based on the clues. If you want to check your fixes, just share them.’ This approach aligned with guided autonomy models, allowing learners to assume increasing responsibility for revisions. A2 learners received more structured guidance that still encouraged cognitive engagement, illustrated by prompts such as ‘Piensa: desayuno ____ la primera clase. ¿Qué preposición usamos para decir ‘durante?’ (‘Think: I have breakfast ____ the first class. Which preposition do we use for ‘during?’). In contrast, A1 learners relied on highly explicit, directive cues, including ‘¿Recuerdas que el pronombre personal ‘él’ lleva tilde?’ (‘Do you remember that the personal pronoun ‘él’ takes an accent mark?’), reflecting reduced opportunities for scaffold fading due to their limited linguistic control.

Rule consolidation (RC). Rule consolidation (RC) similarly scaled with proficiency. B1 learners received reinforcement of abstract grammatical rules after attempting revisions, such

as guidance on tense alignment: ‘To maintain past-tense consistency, you could change it to: ‘I promised ... I’d look more carefully ... ’.’ A2 learners showed partial evidence of rule consolidation, typically at the phrase or morphosyntactic level; for example, ‘You used correctly: ‘we started to work’’. A1 learners’ consolidation was limited to basic morphological adjustments, as when ChatGPT explained: ‘Cambiaste ‘necesito’ por ‘necesita’, ya que hablamos de Pedro’ (‘You changed ‘I need’ to ‘he needs’, because we are talking about Pedro’). This progression from surface-level correction to abstract rule reinforcement reflects growing metalinguistic capacity across proficiency levels.

Metacognitive engagement (ME). Prompts that encouraged learners to explain their reasoning, ask clarifying questions, or reflect on grammatical choices were predominantly observed in the B1 group. ChatGPT frequently invited higher-level reflection, asking questions such as ‘Does the tense match the overall past narrative style? Does it fully convey what you want to say?’ A2 learners demonstrated emerging metacognitive behaviors, often prompted by pragmatic considerations, exemplified by: ‘¿Es apropiado decir que desayunas durante la clase? ¿Qué implicación tiene eso?’ (‘Is it appropriate to say you have breakfast during class? What does that imply?’). At the A1 level, metacognitive engagement was minimal and tended to focus on basic referential clarity rather than strategic reflection, as illustrated by the question ‘¿Quién necesita el checo: Pedro o alguien más?’ (‘Who needs Czech: Pedro or someone else?’). These findings suggest that ME is strongly mediated by linguistic proficiency and becomes more prevalent as learners acquire sufficient resources to monitor and evaluate their own output.

In summary, across all proficiency levels, ChatGPT contributed to reduced cognitive load, supported knowledge retrieval, and facilitated rule awareness. However, the richness of scaffolding, depth of rule consolidation, and level of metacognitive engagement were substantially greater among B1 learners. While A2 learners demonstrated emerging strategic awareness, A1 learners relied heavily on explicit correction cues. These findings indicate that generative AI feedback aligns well with CLT and HCA principles, but its pedagogical impact is mediated by learner proficiency (Figure 2).

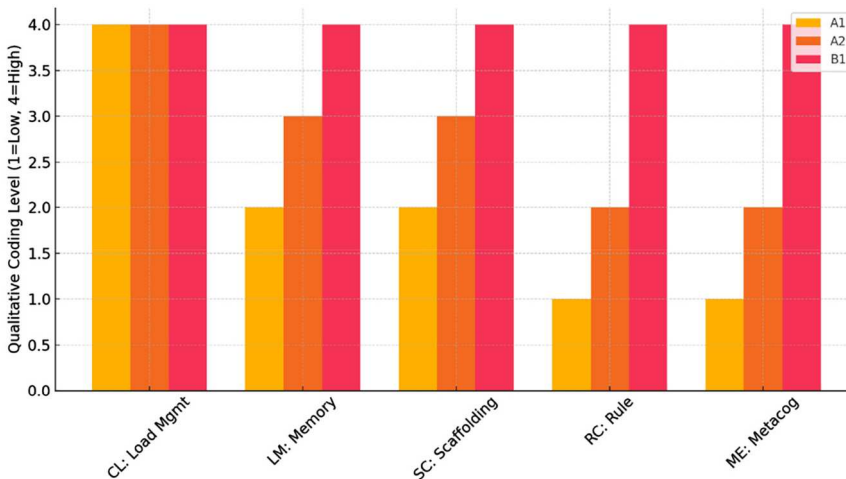


Figure 2. RQ1: Instructional design features across CEFR levels.

RQ2: quality and type of feedback generated by ChatGPT

ChatGPT's feedback was evaluated using Hattie and Timperley's (2007) model, which categorizes feedback according to its instructional focus. The analysis revealed clear differences in the depth and adaptiveness of feedback across proficiency levels.

Task-level feedback (FTL). This was the most prevalent type across all groups. ChatGPT frequently identified surface-level grammatical and lexical errors. For A1 learners, this feedback dominated and often served as a substitute for deeper forms of support.

Process-level feedback (FPL). More complex feedback that addressed patterns or strategies (e.g. 'check your verb endings for agreement') was evident in B1 interactions and occasionally in A2 transcripts. This type of feedback encouraged learners to recognize recurring issues and develop revision strategies.

Self-regulation feedback (FSR). Prompts that asked students to self-assess or reflect on errors were primarily observed in the B1 group. These included indirect cues such as 'Can you spot what might be wrong here?' or 'Try revising this sentence based on the earlier pattern.' Such prompts were rarely present in A1 transcripts.

Self-level feedback (FSL). Praise such as 'Well done!' or 'Good effort' was more common in A1 and A2 transcripts. While potentially motivating, these comments lacked instructional value and were not linked to concrete learning goals.

Overall, ChatGPT provided feedback appropriately matched to learners' proficiency levels, allowing even A1 students to complete the task successfully. This suggests the tool can support lower-level learners effectively, particularly when addressing rule-based, objective errors. At the A1 level, no significant inaccuracies were observed, as the corrections focused on straightforward issues.

In contrast, some B1 transcripts showed that clarification-seeking or open-ended prompts occasionally triggered ambiguous or misleading feedback, particularly when students engaged with more complex language issues such as stylistic nuance or lexical choice. This highlights potential limitations in ChatGPT's reliability for more advanced learners who seek exploratory or nuanced input. Notably, A1 learners, who often rely heavily on AI-generated suggestions, were not exposed to these challenges. This underscores the tool's potential value for supporting beginner-level users.

The quality of feedback was also shaped by how students phrased their prompts and follow-up questions. More targeted or exploratory input typically yielded richer, more adaptive responses, underlining the need for explicit instruction in effective prompt formulation. In summary, ChatGPT's feedback generally aligned with task-level objectives and became more process-oriented as learner proficiency and engagement increased (Figure 3).

5.2. Learner engagement across CEFR levels: metacognitive and affective patterns

The third analysis focused on how learners responded to ChatGPT's scaffolding and feedback, emphasizing cognitive, metacognitive, and affective engagement patterns.

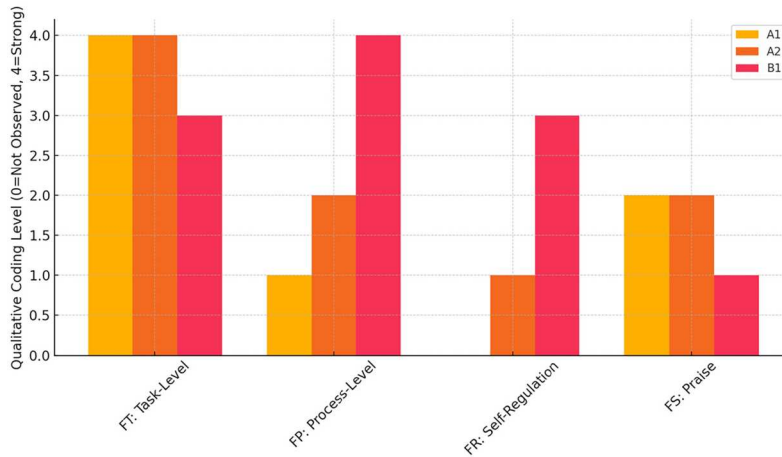


Figure 3. RQ2: Feedback depth across CEFR levels.

RQ3: patterns of engagement across proficiency levels

Learner engagement was evaluated using the ICAP-ME Pedagogical framework. Engagement types were identified in transcript dialogue and post-task reflections, providing a multidimensional view of learner responses. Appendix C includes a sample coded transcript illustrating how these categories were applied in practice.

A1 group engagement profile

The A1 learners primarily demonstrated Passive to Active engagement, with emerging signs of Constructive thinking. Verification Seeking was moderate and focused on confirming sentence-level revisions. Error Awareness emerged in responses to ChatGPT prompts but often lacked deeper explanation. Trust Justification remained mostly Passive, with learners accepting suggestions without critique or reflection. Strategy Talk was modest, mostly indirect or implied. Affective Engagement was present but relatively light, with occasional signs of enjoyment and curiosity (Table 5).

Table 5. A1 Group engagement profile.

Metacognitive Category	Example (Passive)	Example (Active)	Example (Constructive)
Engagement Level(s)			
Verification Seeking (VS) Passive – Constructive	'Is this correct?'	'I applied your correction, is this better?'	'Why was this wrong? Could it be something else?'
Error Awareness (EA) Passive – Constructive	'I noticed it.'	'I fixed the mistake you pointed out.'	'This is wrong because the verb tense doesn't match.'
Trust Justification (TJ) Passive – Active	'I accept this correction.'	'I'm going to look it up to be sure.'	–
Delayed Doubt (DD) Not observed	–	–	–
Strategy Talk (ST) Passive – Active	'I'm trying.'	'Let me try that again.'	–
Affective Engagement (AF) Coded independently from metacognitive behaviors	Example 'This is fun!' 'That surprised me!' 'I'll try it again later.'		
Mildly Positive			

A2 group engagement profile

The A2 learners showed increased metacognitive awareness. Verification Seeking remained common, and Error Awareness involved more independent identification and correction of issues. Trust Justification appeared in the form of critical questioning, and Strategy Talk became clearer through reformulation and self-monitoring. Delayed Doubt was modest but emerging. Affective Engagement remained positive and consistent with prior levels (Table 6).

B1 group engagement profile

The B1 learners demonstrated the richest and most reflective engagement, with frequent Interactive and Constructive metacognitive behavior. Verification Seeking and Error

Table 6. A2 Group engagement profile.

Metacognitive Category Engagement Level(s)	Example (Passive)	Example (Active)	Example (Constructive)	Example (Interactive)
Verification Seeking (VS) Passive – Interactive	'What can I write for that?'	'Are these accents correct? dieciséis, diecisiete, doce, ocho y media.'	'Is it okay to say 'finally to go to bed at eleven?'	One student says, 'That's weird, we did it differently before,' and they discuss it together building shared understanding.
Error Awareness (EA) Passive – Constructive	The learner only corrected the mistake.	'... finally to go to bed at eleven?' – The learner suspects a verb error and checks it.	'I take a shower before sleeping.' – The learner autonomously corrects a reflexive verb structure.	–
Trust Justification (TJ) Active – Interactive	–	'I'll fix it, I prefer the written-out version.'	'This doesn't seem right, because last class we did it differently.' challenges consistency.	One student: 'I read somewhere it's used differently.' ChatGPT replies: 'Really? Where did you see that?' they share reasoning and sources.
Delayed Doubt (DD) Passive – Constructive	Previously incorrect sentence is silently corrected without explanation.	'I fixed it yesterday, I now realize it was wrong.' revisits and updates.	'At first, I thought it was correct, but then I realized I mixed up the prepositions.' reflective insight.	–
Strategy Talk (ST) Passive – Interactive	'I'm not perfect in Spanish. I don't know all the tenses.' – Expresses difficulty without proposing a solution or action.	'Let me try that again.'	'I read the rules first, then answered.'	'Let's try solving the sentence with 'estar' first, then compare.' joint planning and strategy.
Affective Engagement (AF) <i>Coded independently from metacognitive behaviors</i> Mixed (mostly Positive)	Example 'That was cool', 😊, 'It helped me', 'Ugh, this part was confusing' 'This is frustrating, I keep messing it up.' 'Now I understand it better than before.'			

Table 7. B1 group engagement profile.

Metacognitive Category Engagement Level(s)	Example (Passive)	Example (Active)	Example (Constructive)	Example (Interactive)
Verification Seeking (VS) Passive – Interactive	"Is this correct?"	"Should I write 'voy' or 'fui' here?"	"Why can't it be 'por' here? What's wrong with using it?"	Student: 'But I thought 'ser' was for professions?' ChatGPT: 'Yeah, but it's also for characteristics. Let's check.' → They build understanding together.
Error Awareness (EA) Active – Constructive	–	'Oh, I made a mistake, let me fix that.'	'It's wrong because I used the wrong form of 'estar', it's not for permanent states.'	–
Trust Justification (TJ) Passive – Interactive	'Okay, if you say so.'	'I'll check the notes to be sure.'	'But why does it change here? I thought it stays the same with reflexives.' → Challenges the logic of a rule.	Student: 'I think I saw this structure in our workbook.' ChatGPT: 'Oh really? Show me, maybe it helps.' → Offers and builds on a source.
Delayed Doubt (DD) Constructive	–	–	'I originally used 'por' but later realized it should be 'para' because it's a destination.'	–
Strategy Talk (ST) Active – Interactive	–	'I'll rewrite the sentence.'	'First I looked at the verb, then checked the preposition to see if it matched the object.' → Describes a thoughtful strategy.	Student: 'Let's solve the easy parts first, then we'll get the verb right.' ChatGPT: 'Good idea, break it down.' → Joint problem-solving plan.
Affective Engagement (AF) <i>Coded independently from metacognitive behaviors</i> Mixed (curious, positive)	Example 'This is actually kind of fun!' 'Oh wow, I didn't expect that to be wrong.' 'I understand this better than I did in the beginning.'			

Awareness included an analysis of feedback quality. Trust Justification was significantly higher, with learners frequently challenging or defending choices. Delayed Doubt and Strategy Talk both showed strategic re-engagement. Affective Engagement was frequent and reflective, with learners describing insights and learning value (Table 7).

5.2.1. Engagement in post-task reflections

Post-task reflections mirrored transcript findings, validating engagement patterns across CEFR levels. A1 learners emphasized enjoyment and the novelty of using AI. A2 students began articulating revision strategies and describing their reasoning. B1 participants offered evaluative commentary on feedback quality and detailed how specific ChatGPT cues informed their revisions.

Reflections also revealed affective responses, including curiosity, surprise, and a sense of achievement. All learners ultimately identified correct forms with ChatGPT's support, even when initial responses were unclear. Many noted that grappling with uncertainty and asking follow-up questions made learning more memorable. Learners appreciated the ability to request clarification, examples, or translations, and found the interactive format more supportive and personalized than traditional correction tasks.

5.2.2. Summary of RQ3

The three engagement profiles clearly illustrate a developmental trajectory in learner behavior. As proficiency increases, so does the depth of engagement, particularly in metacognitive and strategic dimensions. While affective engagement remained consistently high across all groups, only the B1 learners treated ChatGPT as a collaborative partner, initiating reflective and adaptive interaction. Although the study included participants working in English and Spanish, no meaningful differences were observed in feedback quality or learner engagement (Figure 4).

5.2.3. Linguistic targets and revisions by proficiency level

In addition to metacognitive and affective engagement, transcript analysis revealed consistent patterns in the types of linguistic errors addressed and the revision strategies used, varying by CEFR level and aligning with known developmental trajectories in second language acquisition.

A1 learners focused on basic lexical, morphological, and spelling errors, such as incorrect verb forms, missing accents, and gender agreement. ChatGPT's clues effectively supported correction, and many learners reported that delayed discovery enhanced retention.

A2 learners addressed more complex structures, including reflexives, prepositional phrases, time expressions, and subordination. Their revisions reflected growing syntactic awareness and a shift from isolated correction to integrated language control.

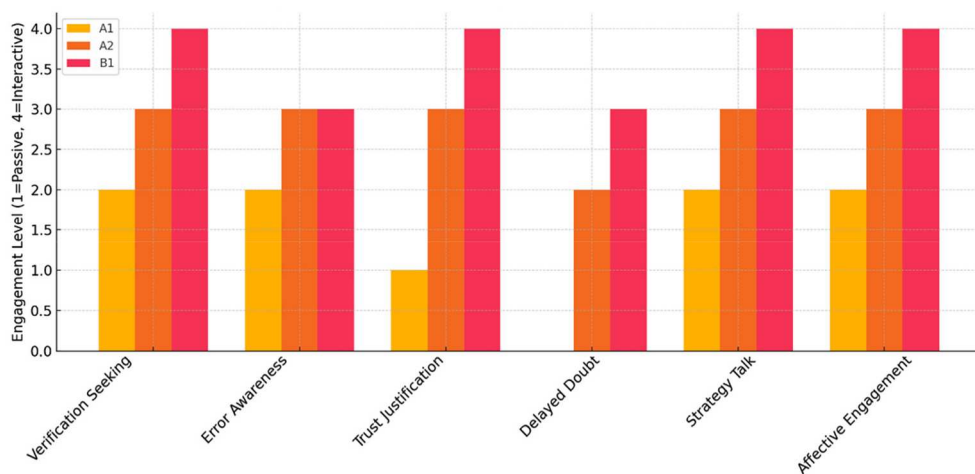


Figure 4. RQ3: Metacognitive and affective engagement across CEFR levels.

B1 learners engaged with advanced features such as tense, aspect, mood, and agreement. They applied rules for progressive and perfective aspect, used the subjunctive appropriately, and revised according to discourse context. These revisions reflected hypothesis testing and consolidation of complex grammar.

Overall, ChatGPT's clue-based feedback supported both surface-level correction and deeper linguistic noticing. Across levels, learners achieved accurate revisions through guided exploration, promoting durable understanding and self-regulated learning.

6. Discussion

This study set out to examine how a clue-based, AI-supported revision task can foster autonomous and reflective language learning across CEFR levels A1- B1, and how learners engage with ChatGPT when required to infer rather than receive corrections. Drawing on Cognitive Load Theory (CLT), Human Cognitive Architecture (HCA), formative feedback models, and the ICAP-ME engagement framework, the findings show that learner proficiency, task design, and AI behavior jointly shape the nature and depth of learning processes.

6.1. *Why B1 learners engaged more constructively*

Across analyzes, B1 learners displayed the most sophisticated metacognitive and interactive engagement. ICAP-ME coding revealed frequent instances of challenging ChatGPT's clues, justifying decisions, calibrating trust, and engaging in strategy-oriented dialogue, behaviors characteristic of constructive and interactive engagement (Chi and Wylie 2014).

From an HCA perspective, these learners possessed the schemas necessary to interpret partial clues without becoming overloaded. The hints functioned as guided discovery supports that activated and refined relevant knowledge structures, consistent with CLT principles (Sweller, Ayres, and Kalyuga 2011). B1 learners also appeared to satisfy autonomy-supportive conditions described in Self-Determination Theory (Deci and Ryan 2000) and research on learner autonomy (Benson 2011). Because they could regulate the timing and degree of help, they engaged in the type of monitoring, verification, and trust calibration central to self-regulated learning (Winne 2018).

By contrast, A1 learners struggled to interpret clues independently. Their limited linguistic resources and still-developing schemas meant that the intrinsic load of constructing meaning in the target language consumed most working-memory capacity. As a result, their behavior skewed toward passive and active ICAP-ME categories, and they relied more heavily on direct feedback. These patterns suggest that clue-based learning aligns well with productive failure and error-based models (Potts and Shanks 2014; Zhang and Fiorella 2022), but only once learners possess enough prior knowledge to benefit from hypothesis generation and refinement.

6.2. *Adaptations for supporting lower-proficiency learners*

Although A1- A2 learners engaged less constructively, the findings do not indicate that the task is unsuitable for beginners. Rather, they highlight the need for targeted

scaffolding aligned with CLT principles to ensure equitable access to metacognitively rich interaction.

Several concrete adaptations emerged:

1. **Prompt Templates and Scaffolds**

Beginners benefit from guided formulations such as ‘Give me a clue about the verb form without correcting it.’ These templates reduce extraneous load and guarantee access to process-level feedback.

2. **Teacher Think-Aloud Modeling**

Demonstrating how to request clues, interpret them, and question AI output externalizes metacognitive strategies (White and Frederiksen 1998) and supports novices in transitioning from passive to active or constructive engagement.

3. **Narrowed Linguistic Scope**

Restricting revision to a small set of features, e.g. past-tense verbs or basic prepositions aligns the cognitive demands of the task with learners’ readiness and reduces overload.

4. **Guided Reflection Questions**

Light reflection prompts such as ‘What mistake did you notice on your own?’ or ‘When were you unsure about ChatGPT’s answer?’ introduce metacognitive monitoring in developmentally appropriate ways.

5. **Collaborative Prompting**

Pair work in designing prompts and interpreting clues distributes cognitive load and supports learning through dialogic interaction, consistent with sociocognitive perspectives on revision (Tulis, Steuer, and Dresel 2017).

These adjustments preserve the logic of productive failure as learners still generate and resolve errors, but they regulate cognitive complexity, making the task accessible across proficiency levels.

6.3. Limitations of ChatGPT’s feedback for advanced learners and pedagogical mitigations

While lower-proficiency learners tended to ask rule-based questions, B1 learners often posed open-ended prompts concerning nuance, style, and complex grammatical choices. These questions occasionally elicited ambiguous or partially inaccurate responses, reflecting well-documented limitations of large language models (Ali et al. 2024; Hua, Jin, and Jiang 2024).

For advanced learners, such inaccuracies have several implications:

- **Misconception and Overgeneralization:** Incorrect or vague feedback can lead learners to consolidate mistaken rules.
- **Distorted Trust Calibration:** Higher fluency may lead to over-trusting the AI, undermining the self-regulatory questioning observed in the ICAP-ME ‘trust justification’ category.
- **Increased Cognitive Load:** Ambiguous clues force learners to reconcile conflicting cues, diverting resources from productive reasoning.

However, the B1 learners in this study frequently counteracted these risks by engaging in verification-seeking, challenging the AI, and triangulating information, behaviors that mitigated the impact of AI inaccuracies. Pedagogical supports can amplify these benefits by:

- Encouraging students to request explanations rather than corrections (Hattie and Timperley 2007).
- Normalizing triangulation with reference materials, corpora, or teacher input.
- Designing tasks that require learners to accept or reject ChatGPT's suggestions and justify their decisions.

In this sense, model inaccuracies, when handled critically, can become opportunities to cultivate AI literacy, reflective judgement, and epistemic vigilance.

6.4. Generalizability across languages and AI systems

The study found no major differences between English and Spanish implementations. However, both are high-resource languages in GenAI training data. Prior research documents substantial English-centric biases in generative models, including more accurate grammar support and richer discourse modeling for English than for less-resourced languages (Brown et al. 2020; Seghier 2023).

Thus, the present findings should not be assumed to generalize to languages with limited training data, non-Latin scripts, or more complex morphological systems. The underlying design principles: error elicitation, clue-based feedback, and guided reasoning, are theoretically language-agnostic, but the reliability, granularity, and pedagogical value of AI feedback may vary widely across languages and models. Broader cross-linguistic and cross-model research is needed to assess performance in typologically diverse contexts.

6.5. Limitations and replicability considerations

Two methodological considerations shape the interpretation and replicability of the findings.

First, variability in learner prompting emerged as a confounding factor. Even with model prompts, differences in phrasing influenced the quality of ChatGPT's responses and subsequent revisions. Because prompting itself is a skill that learners develop over time, it is difficult to determine whether variation in outcomes stemmed from proficiency, prompting ability, or the model's contextual interpretation. Future research should explore ways to train learners in effective prompting or embed automated prompt supports directly into the interface.

Second, while the ICAP-ME framework enabled fine-grained analysis of engagement, some internal processes, such as silent reasoning, hesitation, emotional responses were not fully observable. Post-task reflections provided partial insight, but think-aloud protocols, screen recordings, or interviews could yield richer data on unspoken cognitive and affective processes.

Finally, although engagement patterns were summarized visually, the study's analytical approach was intentionally qualitative. Engagement codes were applied interpretively to capture depth and contextual nuance rather than frequency. In line with design-based and interpretivist research (Hill, Thompson, and Williams 1997; Lincoln and Guba 1985), the goal was to understand how engagement emerged across learner profiles, not to quantify behavioral trends. Mixed-methods approaches could complement these findings by mapping engagement patterns across larger samples.

6.6. Contributions and implications

This study contributes to ongoing efforts to integrate GenAI ethically and pedagogically by demonstrating that the learning value of ChatGPT is shaped not only by model quality but also by task design and learner readiness. By grounding the activity in CLT/HCA and formative feedback theory and analyzing behavior through ICAP-ME, the study offers a conceptual framework for understanding how learners interpret, question, and strategically adapt AI feedback.

The findings show that generative AI can strengthen autonomy, strategic reasoning, and reflective learning, but only when learners possess sufficient prior knowledge or when scaffolding is aligned with cognitive demands. This shifts the conversation beyond 'AI as corrector' toward a more nuanced view of generative AI as a formative learning partner, whose effectiveness depends on the interaction of proficiency, task design, metacognitive strategy, and AI behavior.

7. Conclusion

This study introduces a clue-based revision task that advances AI-mediated language learning by promoting autonomy, strategic thinking, and reflective revision. Its design, grounded in cognitive and feedback theory, enabled learners, especially at higher proficiency levels, to engage with ChatGPT in ways that deepened grammatical understanding and self-regulation.

The task demonstrated how feedback type and quality varied across proficiency levels and depended heavily on how learners formulated their prompts. These findings highlight the importance of training learners in effective prompting and designing AI interactions that support self-regulated learning.

A key contribution is the ICAP-ME pedagogical framework, which extends the original ICAP model by incorporating metacognitive and emotional dimensions of engagement. This framework offers a more comprehensive tool for analyzing learner–AI interaction in real time.

Looking ahead, the task holds strong potential for both formal education and informal, lifelong learning contexts. To fully realize GenAI's educational promise, its use must be intentional, inclusive, and pedagogy-driven. Future research should examine how these approaches perform across languages and learner populations, ensuring that AI-supported learning remains equitable, effective, and adaptable.

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Disclosure statement

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Ethical approval

This research was conducted in accordance with the Ethical Code of the Czech Technical University in Prague, approved by the Ministry of Education, Youth, and Sports under § 36, paragraph 2 of Act No. 111/1998 Coll., on Higher Education Institutions, on April 22, 2022 (file no. MSMT-9911/2022-2). The study also adhered to the ethical guidelines outlined in the Declaration of Helsinki.

University student participants were fully informed about the nature and scope of the research and participated voluntarily. Informed consent was obtained from all participants, and they were free to withdraw at any point without consequence. Data were pseudonymized and securely stored in accordance with institutional and GDPR guidelines. No sensitive personal data were collected.

Informed consent

All participants were thoroughly informed about the research and gave consent to the use of the data collected.

Data availability statement

The datasets generated and analyzed during the current study, including the raw data supporting the findings, are available from the corresponding author upon reasonable request and within the article; however, some data cannot be provided to protect the students' anonymity.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT for language editing and formatting references. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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